

# Vinu K. Vijayakumar

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## PROFESSIONAL SUMMARY

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Senior Data Scientist with 3+ years of experience in digital lending and fintech, specializing in credit risk modeling, credit policy management, and AI-powered automation. Proven track record of developing 8 ML models with 6 deployed to production, driving 10 major credit policy overhauls, and delivering measurable portfolio risk improvements across unsecured business lending. Experienced in cross-functional collaboration, model validation, and translating complex data into actionable credit strategies.

## WORK EXPERIENCE

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**Protium Finance Ltd** | Digital Lending · Business Lending · Fintech

### **Senior Data Scientist** *Mar 2024 – Present*

- Led and mentored a 4-6 member engineering team to enhance a production Business Rules Engine (BRE) — the platform underpinning 74 credit policies (10 major overhauls) across unsecured digital business lending — enabling version-controlled policy deployments, faster release cycles, and scalable policy experimentation in production.
- Led cross-functional discussions for credit decisioning platform enhancements, translating business requirements into technical solutions, assessing downstream system impacts, and driving execution across product, risk, and engineering teams.
- Led end-to-end customer risk segmentation using decision trees across bureau, banking, and loan-level dimensions — identifying hard policy boundary conditions and producing a risk stratification framework that became the foundation of the next generation of credit policies, improving overall portfolio risk metrics by ~20%.
- Designed, built, and deployed a production service for bureau and banking PDF parsing, leveraging AI-assisted software development workflows to accelerate implementation, testing, and deployment.
- Architected modular dependency management for the banking parser by isolating OCR and ePDF pipelines into optional install paths, enabling lightweight deployments for legacy systems; resolved UAT deployment issues through Docker-based debugging and environment stabilization.
- Developed LLM-powered automated validation modules for datasets and Jupyter notebooks using LiteLLM, automatically detecting data quality issues, logical inconsistencies, and code-level defects.
- Built autonomous AI agents for code review and automated test generation to improve production code quality and reduce manual validation effort.

### **Data Scientist** *Jan 2023 – Mar 2024*

- Developed 8 production-grade credit risk scorecards and ML models, with 6 deployed to production across various periods — spanning 2 Experian-based scorecards, 3 CIBIL-based scorecards, 1 banking-data scorecard, and 2 multi-bureau models (CIBIL + Experian + banking) — built using XGBoost, gradient boosting, Weight of Evidence (WOE)-based Logistic Regression, and Reject Inference, achieving 5-10% AUC improvement while reducing portfolio risk exposure.
- Validated and documented both secured and unsecured credit risk and approval models, ensuring regulatory compliance and robust statistical performance prior to production roll out.

- Built and productionized a propensity-based customer ranking ML model to prioritize high-conversion loan applicants, improving targeted disbursal effectiveness by 12%; implemented a champion-challenger framework enabling simultaneous live routing of different customer cohorts to different policy versions, with the champion policy variant outperforming the challenger by 5%, validating the deployment decision.
- Developed a data-driven partner pre-approval selection framework — a statistical customer eligibility scoring process delivered to lending partners — improving their loan approval rates by ~30% and expanding the disbursable pipeline.
- Reduced high-volume batch data pipeline runtime from 2 hours to 10 minutes through parallelized, optimized data processing and scrubbing workflows.
- Designed scalable bureau data parsing and feature engineering systems generating 30+ core features with 1000+ permutations, reducing storage usage by 70% and improving feature computation speed by 10x.
- Conducted exploratory analysis and segmentation experiments to support credit approval optimization and inform policy design decisions.
- Performed load testing on credit decisioning systems during policy deployments, ensuring platform stability under peak lead flow.

## EDUCATION

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### Indian Institute of Science Education and Research (IISER), Thiruvananthapuram

BS-MS Dual Degree (Master's equivalent) | Physics with Mathematics Minor | Aug 2015 – Jun 2020

**Thesis:** *Detection of Chemicals via Surface-Enhanced Raman Spectroscopy*

## SKILLS

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**Programming:** Python (Pandas, NumPy, Scikit-learn, FastAPI, Locust), SQL, Git

**Machine Learning:** Credit Risk Modeling, Logistic Regression, XGBoost, Gradient Boosting, Reject Inference, Scorecard Development, Feature Engineering, Feature Selection, Model Deployment (CI/CD), Model Monitoring, Model Validation

**Statistical Analysis:** Regression Analysis, Statistical Modeling, Champion-Challenger Testing, Weight of Evidence (WOE), Information Value (IV)

**Credit and Risk:** Credit Scoring, Credit Policy Management, Portfolio Risk Management, Risk Segmentation, Auto-Approval Programs, Adverse Selection, Bureau Analysis (CIBIL, Experian)

**Data Engineering:** Parallelized Data Processing, Data Scrubbing, Bureau and Banking Data Parsing, Docker, AWS, Athena

**AI and Automation:** LLM Applications, LiteLLM, AI Agents, Automated Validation Systems, NLP

**Domain:** Digital Lending, Business Lending, Unsecured Credit, Fintech, Credit Decisioning Platforms

**Tools:** PyCharm, DBeaver, Excel, Opencode, Goose, Claude Code